**SET-1**

**Series A6BAB/C**

प्रश्न-पत्र कोड
Q.P. Code **56/6/1**

रोल नं.
Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **11** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **12** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **11** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **12** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

# रसायन विज्ञान (सैद्धान्तिक)

# CHEMISTRY (Theory)

*निर्धारित समय : 2 घण्टे* — *अधिकतम अंक : 35*

*Time allowed : 2 hours* — *Maximum Marks : 35*

***General Instructions :***

*Read the following instructions very carefully and strictly follow them :*

*(i)* *This question paper contains **12** questions. **All** questions are compulsory.*

*(ii)* *This question paper is divided into **three** Sections – **A**, **B** and **C**.*

*(iii)* ***Section A** – Questions no. **1** to **3** are very short answer type questions, carrying **2** marks each.*

*(iv)* ***Section B** – Questions no. **4** to **11** are short answer type questions, carrying **3** marks each.*

*(v)* ***Section C** – Question no. **12** is case based question, carrying **5** marks.*

*(vi)* *Use of log tables and calculators is **not** allowed.*

## SECTION A

**1.** Answer the following questions (Any ***two***) : *2×1=2*

(a) Why is alternating current used instead of direct current for measuring resistance of an electrolytic solution ?

(b) State Kohlrausch's law of independent migration of ions.

(c) Depict the Galvanic cell in which the following reaction takes place :

$$Mg\,(s) + Cu^{2+}\,(aq) \longrightarrow Mg^{2+}\,(aq) + Cu\,(s)$$

**2.** (a) For a reaction $R \rightarrow P$, half life ($t_{1/2}$) is independent of the initial concentration of reactants. What is the order of the reaction ?

(b) Write one difference between order and molecularity of a reaction. *2×1=2*

**3.** (a) Arrange the following compounds in the increasing order of their acidic strength :

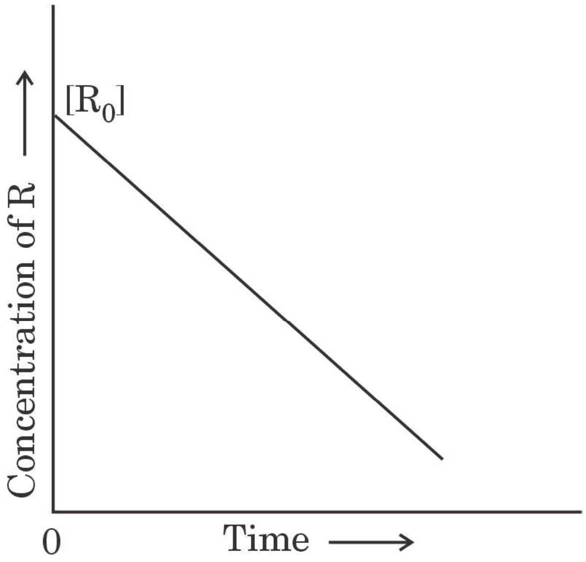
$$Cl - CH_2 - COOH, \quad \underset{\displaystyle Cl}{\underset{|}{Cl - CH}} - COOH, \quad CCl_3 - COOH$$

(b) Write the IUPAC name of the given compound :

$CH_3COCH_2COCH_3$ *2×1=2*

## SECTION B

**4.** (a) The variation in the concentration (R) vs. time (t) plot is given below. Answer the following questions on the basis of the given figure : *3×1=3*




(i) Predict the order of the reaction.

(ii) What is the slope of the curve ?

(iii) What are the units of the rate constant k ?

**OR**

(b) A first order reaction takes 24 minutes for 50% decomposition. Calculate the time required for 25% decomposition. *3*

(Given : log 4 = 0·6021, log 3 = 0·4771)

**5.** (a) (i) Which is the most common oxidation state in Lanthanoids ?

(ii) Why is there a gradual decrease in the atomic sizes of transition metals in a series with increasing atomic numbers ?

(iii) Calculate the number of unpaired electrons in the following gaseous ions : *3×1=3*

$V^{3+}$, $Ti^{3+}$

**OR**

(b) (i) The electronic configuration of Ce is :

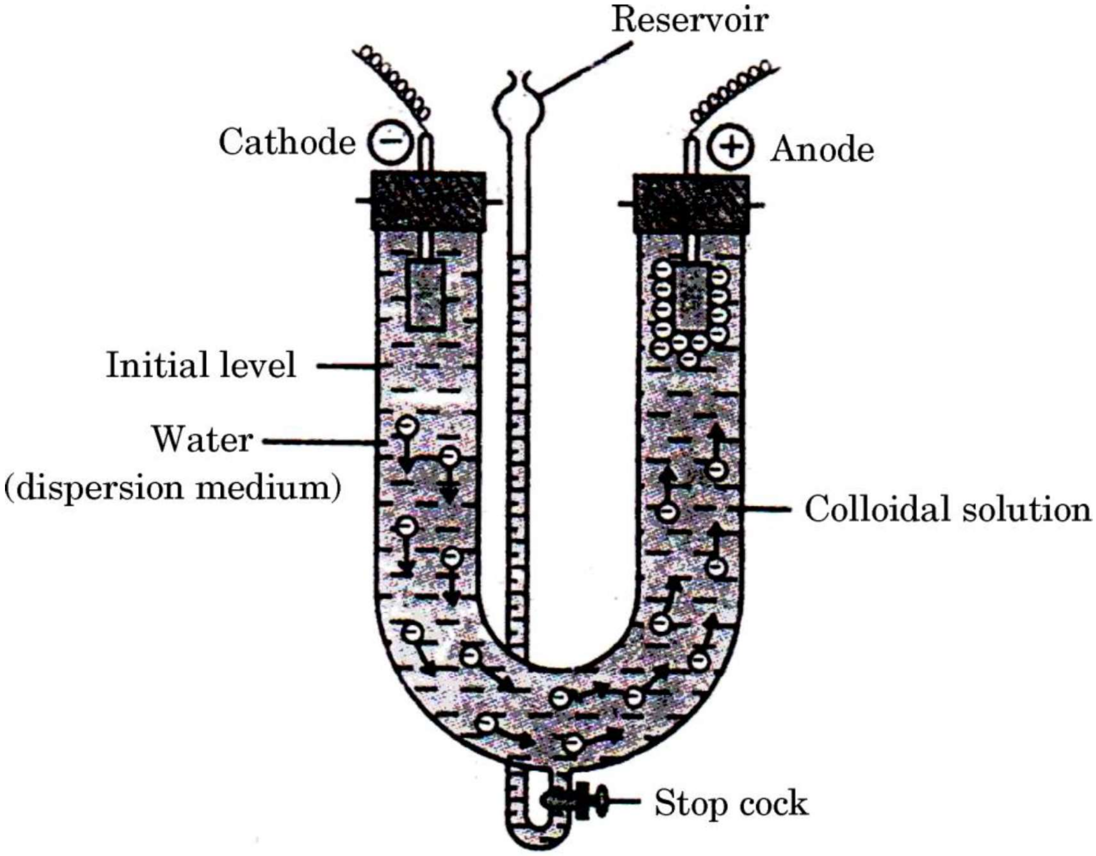
$$_{58}Ce = [Xe]^{54}\ 4f^1\ 5d^1\ 6s^2$$

Calculate the spin only magnetic moment of $Ce^{3+}$ ion.

(ii) Why is copper regarded as a transition element although copper atom has completely filled d-orbitals in its ground state ?

(iii) Why is $Sc^{3+}$ colourless in aqueous solution whereas $Ti^{3+}$ is coloured ? *3×1=3*

**6.** Answer the following questions on the basis of the given figure : *3×1=3*




(a) Define the process depicted in the above figure.

(b) Can this process be used in the coagulation of the lyophobic sols ?

(c) What is coagulation ?

**OR**

Define the following terms : *3×1=3*

(a) Adsorption

(b) Lyophobic sol

(c) Multimolecular colloid

**7.** The conductivity of $0{\cdot}001$ mol $L^{-1}$ solution of acetic acid is $3{\cdot}905 \times 10^{-5}$ S $cm^{-1}$. Calculate its molar conductivity. If $\Lambda_m^o$ for acetic acid is $390{\cdot}5$ S $cm^2$ $mol^{-1}$, then calculate its degree of dissociation ($\alpha$). *3*

8. (a) Choose the primary amine from the following compounds : *1*

$(C_2H_5)_2CHNH_2$, $(C_2H_5)_2NH$, $(CH_3)_3N$

(b) Complete the following : *2*

$$CH_3Br \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B$$

9. (a) Write the hybridisation, shape and magnetic character of $[CoF_6]^{3-}$.
[Atomic number of Co = 27]

(b) Write the formula of Pentaamminechloridocobalt(III)chloride. *2+1=3*

10. Account for the following : *3×1=3*

(a) Transition elements show variable oxidation states.

(b) $E^o_{Cu^{2+}/Cu}$ value for copper is highly positive.

(c) $Cr^{2+}$ is a strong reducing agent.

**OR**

What is Lanthanoid contraction ? Write its two consequences. *3*

11. (a) Aromatic primary amines cannot be prepared by Gabriel Phthalimide synthesis. Why ?

(b) Aniline is a weaker base than alkyl amines. Why ?

(c) Arrange the following in the increasing order of basic strength in aqueous solution :

$(CH_3CH_2)_2NH$, $(C_2H_5)_3N$, $C_2H_5NH_2$ *3×1=3*

## SECTION C

12. Read the following passage and answer the questions that follow : *1+1+1+2=5*

A class of organic molecules which contain a carbon atom connected to an oxygen atom by a double bond is called Aldehydes and Ketones. It is called as carbonyl group. Aldehydes are prepared by the oxidation of alcohols. Formaldehyde is sold in an aqueous solution called formalin. Propanone, a simplest ketone is commercially prepared by fermenting corn or by oxidation of propan-2-ol. Carboxylic acids also have carbonyl carbon. They can be prepared by the oxidation of alcohols and aldehydes.

Formic acid was first isolated by the distillation of red ants. It is partially responsible for the pain and irritation of ant and wasp stings. Aldehydes undergo many nucleophilic addition reactions. They can be reduced to primary alcohols. The aldehydes with $\alpha$-hydrogen undergo aldol condensation and the aldehydes without $\alpha$-hydrogen undergo Cannizzaro reaction.

Ketones are highly reactive, although less so than aldehydes. Ketones are obtained by oxidation of secondary alcohols. Ketones possessing $\alpha$-hydrogens also undergo aldol condensation. Carboxylic acids occur widely in nature and are used in the production of plastics, esters, etc. Aspirin is prepared from acetic acid. Similar to aldehydes and ketones, carboxylic acids can be halogenated at the $\alpha$-carbon by reacting with a halogen in presence of phosphorus.

(a) Which of the following compounds would undergo aldol condensation ?

Methanal, Benzaldehyde, Ethanal

(b) Write the chemical test to distinguish between propanal and propanone.

(c) Write the reagent required in the following reaction :

$$CH_2 = CH - CH_2OH \xrightarrow{?} CH_2 = CH - CHO$$

(d) (i) An alcohol 'A', ($C_3H_8O$) on oxidation gives compound 'B'. 'B' gives negative Tollens' test and reacts with hydrazine to give compound 'C'. 'B' reacts with NaOH and $I_2$ to give yellow precipitate of 'D'. Identify 'A', 'B', 'C' and 'D'.

**OR**

(ii) Write the chemical reactions for the following :

(I) Clemmensen reduction

(II) HVZ reaction